

Comparative Recommendation Engineering Framework & Model Benchmarking

Academic Evaluation Presentation — Multi-Paradigm Recommendation System Modeling & SHAP Explainability Architecture.

99,287

Ratings Evaluated
S-Core pruned interactions

943

Users
Unique user profiles

1,349

Items
Film catalogue

7.80%

Density
Interaction matrix sparsity



Feature Store Architecture

78 Multi-modal features structured across 4 specialized representation categories

CATEGORY 01

7 Features

Popularity & Biases

- ✓ user_bias
- ✓ item_bias
- ✓ item_count
- ✓ user_mean_rating

CATEGORY 02

33 Features

Latent Factors

Low-rank matrix factorization capturing implicit feedback & user-item interaction geometry.

- 33 SVD matrix factorization features

CATEGORY 03

19 Features

Metadata & Genres

- Action
- Sci-Fi
- Comedy
- movie_age_years

CATEGORY 04

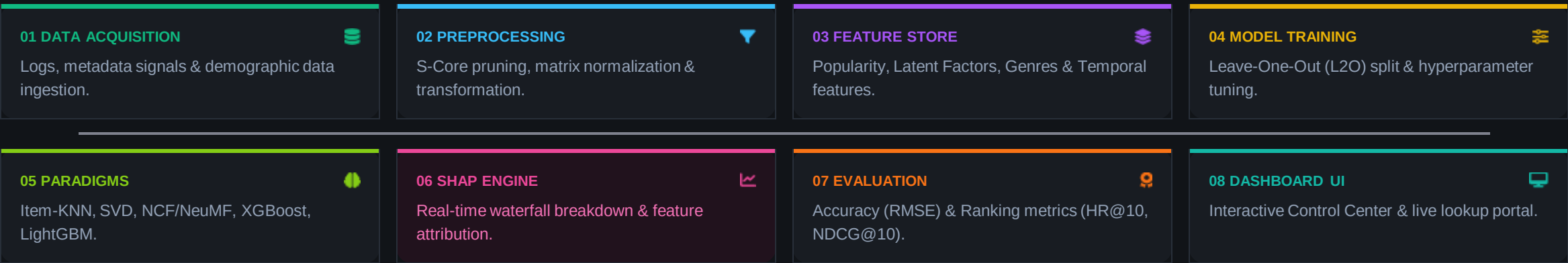
Temporal

Temporal Signals

- 🕒 day_of_week
- 🕒 hour
- 🕒 user_tenure_days
- 🕒 year

End-to-End System Design & Data Flow

Modular recommendation engineering framework from data ingestion to XAI attribution

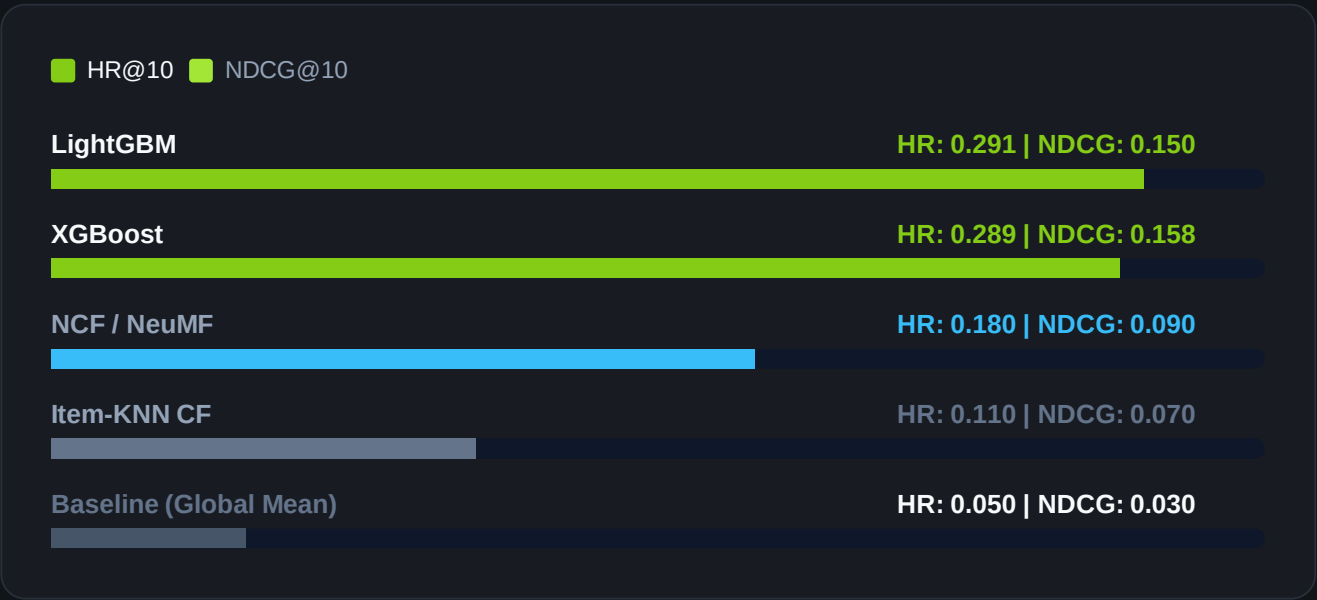


Core Pipeline Invariant: Closed-loop feedback connects XAI feature attribution directly into hyperparameter tuning.

SHAP XAI ENGINE HIGHLIGHTED

Model Evaluation & Performance Comparison

Multi-paradigm ranking performance evaluation on Hit Rate (HR@10) and NDCG@10



🏆 Gradient Boosting Dominance

Tree-based ensemble models (XGBoost / LightGBM) significantly outperform all other paradigms in both rating accuracy and ranking quality.

⚡ **HR@10 $\approx$ 0.29:** Nearly 3 $\times$ improvement over traditional Item-KNN CF.

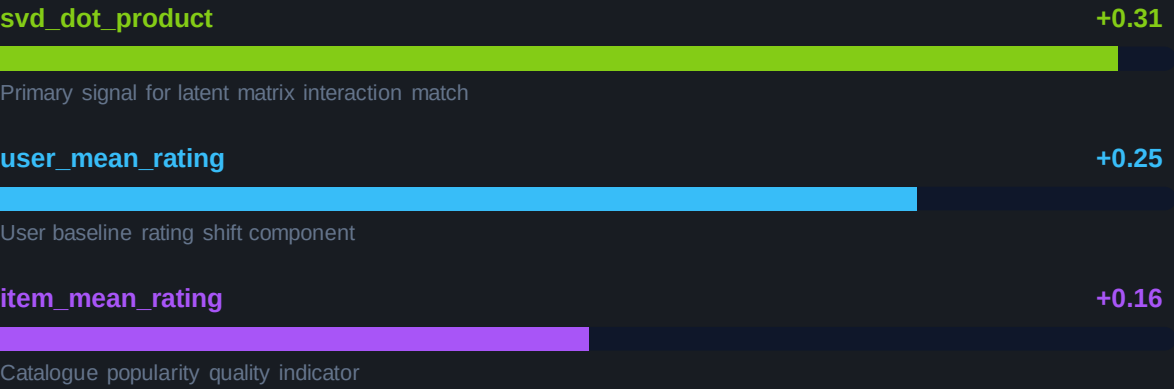
📈 **NDCG@10 $\approx$ 0.15–0.16:** Superior ranked relevance across user interaction trajectories.

🧱 **Feature Value:** Engineered tabular features overcome baseline cold-start barriers.

Search & Explainable Recommendation Engine

Global and local feature attribution utilizing SHAP (Shapley Additive exPlanations)

Global Model Feature Attribution (Mean |SHAP| Value)



LOCAL EXPLANATIONS

Real-time SHAP Waterfalls

Decomposes individual user–item predictions into additive feature contributions, exposing positive vs. negative drivers.

GLOBAL BEHAVIOR

Model Governance

Aggregated test SHAP distributions validate latent factorization signals as the dominant predictor across all user groups.

Interactive Control Center & Research Roadmap

Web UI capabilities and future milestone evaluation plan

Control Center Web UI Capabilities

Dataset Analytics

Quality metrics, transformation logs & density visualizations.

Feature Store Preview

Variable distribution plotting across all 78 engineered features.

Live SHAP Explanations

Interactive User ID lookup with real-time waterfall attribution plots.

Research Paper Draft Status

01 Methodology & Architecture

Pipeline & model paradigms documented

DONE

02 Benchmark Results

Comparative evaluation tables compiled

DONE

03 Real-Time Serving Endpoint

Low-latency inference API deployment

PLANNED

04 Scale-Up Validation

Extend to MovieLens 1M & 20M datasets

PLANNED